

Homework 4 Solutions

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Problem 9.1

$$720 = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 6!.$$

$$n = 6.$$

Problem 9.2

- a) This is equivalent to asking "how many ways can 19 books be arranged?", and so the answer is $19!$.
- b) We have $6!$ ways of arranging the French books, $8!$ ways of arranging the Russian books, $5!$ ways of arranging the Spanish books, and $3!$ ways of arranging the groups of books. Therefore the answer is $6! \cdot 8! \cdot 5! \cdot 3!$.

Problem 9.4

Proof. If $k, n \in \mathbb{N}$ such that $k \leq n$, then $(n)_k = \frac{n!}{(n-k)!}$

1. Let $k, n \in \mathbb{N}$ such that $k \leq n$.

$$2. \text{ Then } \frac{n!}{(n-k)!} = \frac{n(n-1)(n-2)\dots(n-k+1)(n-k)(n-k-1)\dots(2)(1)}{(n-k)(n-k-1)\dots(2)(1)} = n(n-1)(n-2)\dots(n-k+1) = (n)_k.$$

$$3. \text{ Therefore } (n)_k = \frac{n!}{(n-k)!}.$$

□

Problem 9.5

$$\frac{100 \cdot 99 \cdot 98 \cdot 97 \cdot \dots \cdot 1}{98 \cdot 97 \cdot 96 \cdot 95 \cdot \dots \cdot 1} = 100 \cdot 99 = 9900.$$

Problem 9.9

- a) 1
- b) 5
- c) 23
- d) 119
- e) 719

$$\sum_{k=1}^n k \times k! = (k+1)! - 1$$

Problem 9.10

24 zeros (one for every multiple of 5 and an extra one for multiples of 25).

Problem 9.11

Proof. The list of numbers $1000! + 2, 1000! + 3, 1000! + 4, \dots, 1000! + 1002$ are composite.

1. Let $A = \{1000! + 2, 1000! + 3, 1000! + 4, \dots, 1000! + 1002\}$.
2. Let $x \in A$, and observe that $\exists y \in \mathbb{N}$ such that $x = 1000! + y$.
3. Observe that y is an integer such that $2 \leq y \leq 1002$. (Definition of set A)
4. Then $x = y(\frac{1000!}{y} + 1)$. (Distribution)
5. Observe that $\frac{1000!}{y}$ is an integer for all $y \leq 1000$. (Integers are closed under multiplication)
6. Observe that for $y = 1001$, y can be expressed as the unique factors 77×13 , showing that $\frac{1000!}{1001}$ is an integer. (Integers are closed under multiplication)
7. Observe that for $y = 1002$, y can be expressed as the unique factors 2×501 , showing that $\frac{1000!}{1002}$ is an integer. (Integers are closed under multiplication)
8. Therefore $\frac{1000!}{y}$ is an integer for all values of y . (Result of 5, 6, and 7)
9. There exists an integer a such that $ya = x$, namely $a = \frac{1000!}{y} + 1$, showing that $y|x$. (Definition of divisible)
10. Observe that $1 < y$. (Definition of y)
11. Observe that since $0 < 1000!$, then $y + 0 < y + 1000!$ and therefore $y < x$. (Addition and substitution)
12. Therefore $1 < y < x$ and $y|x$. (Result of 8, 9, and 10)
13. Therefore x is composite. (Definition of composite)
14. Therefore every element of A is a composite number. (Definition of x) □

Problem 10.1

- a) $\{0, 3, 6, 9\}$
- b) $\{2\}$
- c) $\{-2, 2\}$
- d) $\emptyset$
- e) $\{\emptyset\}$
- f) $\{-10, 10, -20, 20, -50, 50, -100, 100\}$
- g) $\{\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{5\}\}$

Problem 10.2

- a) $\{x \in \mathbb{N} : 0 < x \leq 10\}$
- b) $\{x \in \mathbb{Z} : -8 \leq x \leq 8\}$
- c) $\{x \in \mathbb{N} : x \text{ is odd}\}$
- d) $\{x : x = y^2 \text{ where } y \in \mathbb{N} \text{ and } 0 < y \leq 10\}$

Problem 10.3

- a) 21
- b) 2
- c) 0
- d) 0
- e) ∞
- f) 256
- g) 4
- h) 2

Problem 10.4

- a) $\in$
- b) $\subseteq$
- c) $\in$
- d) $\subseteq$
- e) $\subseteq$
- f) $\subseteq$
- g) $\in$

Problem 10.5

- a) $A = \{1\}, B = \{1\}, C = \{1\}$
- b) $A = 1, B = \{1\}, C = \{1\}$
- c) $A = 1, B = \{1\}, C = \{\{1\}\}$
- d) $A = \{1\}, B = \{1\}, C = \{\{1\}\}$

Problem 10.6

- a) $A = \{1\}$
- b) $A = \{\emptyset\}$
- c) $A = \emptyset$
- d) No solution. $\emptyset$ has no elements.

Problem 10.7

- a) *Proof.* If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$
 - 1. Let A, B , and C be sets such that $A \subseteq B$ and $B \subseteq C$.
 - 2. Let $x \in A$.
 - 3. Since $A \subseteq B$ then $x \in B$. (Definition of subset)
 - 4. Since $B \subseteq C$ then $x \in C$. (Definition of subset)
 - 5. Therefore, $x \in A$ implies $x \in C$. 6. Therefore, $A \subseteq C$. (Definition of subset)
 - 7. Therefore, if $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$. □
- b) *Proof.* If $A \in B$ and $B \subseteq C$, then $A \subseteq C$ is *false*.
 - 1. Let $A = 1$, let $B = \{1\}$, and let $C = \{1\}$.
 - 2. Observe that $A \in B$ and $B \subseteq C$, but A is not a subset of C . □
- c) *Proof.* If $A \in B$ and $B \subseteq C$, then $A \in C$.
 - 1. Let A, B , and C exist such that $A \in B$ and $B \subseteq C$.
 - 2. Let $x \in B$.
 - 3. Observe that since $x \in B$, then there exists some x such that x is the element A . (Definition of x)
 - 4. Since $B \subseteq C$ then for all $x, x \in C$. (Definition of subset)
 - 5. Since there exists some x such that x is the element A and since for all $x, x \in C$, then $A \in C$. (Result of 3 and 4)
 - 5. Therefore, if $A \in B$ and $B \subseteq C$, then $A \in C$. □

d) *Proof.* If $A \in B$ and $B \in C$, then $A \in C$ is not always true.

1. Let $A = 1$, $B = \{1\}$, and $C = \{\{1\}\}$.
2. Observe that $A \in B$ and $B \in C$.
3. However, $A \in C$ is false.
4. Therefore if $A \in B$ and $B \in C$, then $A \in C$ is not always true. □

Problem 10.8

Proof. $A = B$ iff $A \subseteq B$ and $B \subseteq A$.

1. ($\Rightarrow$) Let A and B be sets such that $A = B$.
2. Then $A \subseteq A = B$ gives $A \subseteq B$ and also $B \subseteq B = A$ gives $B \subseteq A$.
3. ($\Leftarrow$) Let A and B be sets such that $A \subseteq B$ and $B \subseteq A$.
4. Let $x \in A$, and observe that since $A \subseteq B$ then $x \in B$.
5. Next, let $x \in B$, and observe that since $B \subseteq A$ then $x \in A$.
6. Therefore $A = B$.
7. Therefore $A = B$ iff $A \subseteq B$ and $B \subseteq A$. □

Problem 10.9

Proof. If $A \subseteq B$, $B \subseteq C$, and $C \subseteq A$, then $A = C$

1. Let A , B , and C be sets such that $A \subseteq B$, $B \subseteq C$, and $C \subseteq A$.
2. Let $x \in A$.
3. Since $A \subseteq B$ then $x \in B$, and since $B \subseteq C$ then $x \in C$.
4. Now let $x \in C$.
5. Since $C \subseteq A$ then $x \in A$.
6. Therefore $A = C$ □

Problem 10.10

Proof. If $A = \{x \in \mathbb{Z} : 4|x\}$ and $B = \{y \in \mathbb{Z} : 2|y\}$ then $A \subseteq B$

1. Let $A = \{x \in \mathbb{Z} : 4|x\}$ and $B = \{y \in \mathbb{Z} : 2|y\}$.
2. Let $x \in A$, and observe that $4|x$. (Definition of A)
3. Since $4|x$, then there exists some integer a such that $4a = x$. (Definition of divisible)
4. Then $x = 4a = 2(2a)$. (Distribution)
5. Then there exists some integer b such that $2b = x$, namely $b = 2a$, showing that $2|x$. (Definition of divisible)
6. Since $2|x$, then $x \in B$. (Definition of B)
7. Therefore every element of A is also an element of B showing that $A \subseteq B$. (Definition of subset)
8. Therefore if $A = \{x \in \mathbb{Z} : 4|x\}$ and $B = \{y \in \mathbb{Z} : 2|y\}$, then $A \subseteq B$. □

Problem 11.1

- a) $\forall x \in \mathbb{Z}, x$ is prime.
- b) $\exists x \in \mathbb{Z}, x$ is not prime and x is not composite.
- c) $\exists x \in \mathbb{Z}, x^2 = 2$.
- d) $\forall x \in \mathbb{Z}, 5|x$.
- e) $\exists x \in \mathbb{Z}, 7|x$.
- f) $\forall x \in \mathbb{Z}, x^2 \geq 0$.
- g) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, xy = 1$.
- h) $\exists x \in \mathbb{Z}, \exists y \in \mathbb{Z}, \frac{x}{y} = 10$.
- i) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, xy = 0$.

- j) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, y > x$.
- k) $\forall x \in \text{People}, \exists y \in \text{People}, x \text{ loves } y \text{ for some time}$.

Problem 11.2

- a) $\exists x \in \mathbb{Z}, x \text{ is not prime}$.
There exists an integer that is not prime.
- b) $\forall x \in \mathbb{Z}, x \text{ is prime and } x \text{ is composite}$.
All integers are prime and composite.
- c) $\forall x \in \mathbb{Z}, x^2 \neq 2$.
All integers when squared are not equal to 2.
- d) $\exists x \in \mathbb{Z}, 5 \nmid x$.
There exists an integer that is not divisible by 5.
- e) $\forall x \in \mathbb{Z}, 7 \nmid x$.
All integers are not divisible by 7.
- f) $\exists x \in \mathbb{Z}, x^2 < 0$.
There exists an integer whose square is less than 0.
- g) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, xy \neq 1$.
There exists an integer that for all other integers, their product is not 1.
- h) $\forall x \in \mathbb{Z}, \forall y \in \mathbb{Z}, \frac{x}{y} \neq 10$.
For any two integers, their quotient is not 10.
- i) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, xy \neq 0$.
For all integers, there exists another integer such that their product is not 0.
- j) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, y \leq x$.
There exists an integer such that all other integers are less than or equal to it.
- k) $\exists x \in \text{People}, \forall y \in \text{People}, x \text{ never loves } y \text{ for some time}$.
Somebody has never loved anybody sometime.

Problem 11.3

"Everyone is not invited to my party" means " $\forall x \in \text{People}, x \text{ is not invited to my party}$ ". Presumably, they intended "Not everyone is invited to my party", or " $\exists x \in \text{People}, x \text{ is not invited to my party}$ ".

Problem 11.4

- a) False
- b) True
- c) False
- d) True
- e) False
- f) True
- g) True
- h) True

Problem 11.5

- a) $\exists x \in \mathbb{Z}, \neg(x < 0)$
There exists an integer that is not less than zero.
- b) $\forall x \in \mathbb{Z}, \neg(x = x + 1)$
For all integers, an integer is not equal to itself plus one.
- c) $\forall x \in \mathbb{N}, \neg(x > 10)$.
All natural numbers are not greater than 10.
- d) $\exists x \in \mathbb{N}, \neg(x + x = 2x)$
There exists a natural number such that when added with itself, the sum does not equal twice itself.
- e) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, \neg(x > y)$
For all integers, there exists a second integer such that the second integer is not less than the first.
- f) $\exists x \in \mathbb{Z}, \exists y \in \mathbb{Z}, \neg(x = y)$
There exists an integer, such that there exists a second integer where the second integer is not equal to the first.
- g) $\exists x \in \mathbb{Z}, \forall y \in \mathbb{Z}, \neg(x + y = 0)$
There exists an integer such that for all other integers, their sum is not 0.

Problem 11.6

Note if we let S represent some assertion about some variable, then "some assertion about x and y " can be symbolized as $S(x) \wedge S(y)$. Then the statements " $\forall x, \forall y, S(x) \wedge S(y)$ " is logically equivalent to " $\forall y, \forall x, S(x) \wedge S(y)$ " since " $\wedge$ " is commutative and x and y are just dummy variables. In other words " $\forall x, \forall y, S(x) \wedge S(y)$ " = " $\forall y, \forall x, S(y) \wedge S(x)$ " by just switching the names of the dummy variables, and then " $\forall y, \forall x, S(y) \wedge S(x)$ " = " $\forall y, \forall x, S(x) \wedge S(y)$ " since " $\wedge$ " is commutative. Similar argument for the statements " $\exists x, \exists y, S(x) \wedge S(y)$ " and " $\exists y, \exists x, S(x) \wedge S(y)$ ".

Problem 11.7

- a) True, 2.
- b) False, both 2 and -2.
- c) False, square root of three is not a natural number.
- d) True, 0
- e) True, 1